Data Sheet

EXPERIMENT NO. 12

Title: - Simulation of Schmitt Trigger Circuit using IC 555 (using Tinkercad / Proteus software circuit).

Instrument & Component Required: -

Sl. No.	Apparatus / Component	Specification	Quantity
1	Timer IC	555	1
2	Breadboard	Small	1
3	DC Power Supply	10V	1
4	Function Generator	1 KHz, 2V	1
5	Oscilloscope	–	1
6	Multimeter	Digital	1
7	Resistor	–	required
8	Capacitor	–	required.

Important Formulae: -

Operating Principle: -

Pins 2 and 6 of the IC 555 are tied together and driven by the input signal. The internal comparators (with reference levels at V_{CC} /3 and 2 V_{CC} /3) convert a slowly-varying input into a clean rectangular output with two well-defined switching thresholds, giving the circuit hysteresis.

Upper & Lower Threshold Voltages: -

Upper Threshold Voltage, $V_{UT} = (2/3) \times V_{CC}$

Lower Threshold Voltage, $V_{LT} = (1/3) \times V_{CC}$

Hysteresis Width: -

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Title:- Simulation of Schmitt trigger Circuit using IC555 (Using TinkerCad / Proteus).

Theory:-

The Schmitt Trigger Circuit using an IC 555 is designed to convert a slowly varying input signal into a clean, rectangular output waveform. This is achieved by tying pins 2 and 6 of the IC 555 together and driving them directly with the input signal.

① Upper Threshold Voltage:

$V_{ut} = (2/3) \times V_{cc}$

② Lower Threshold voltage:

$V_{LT} = (1/3) \times V_{cc}$

③ Hysteresis Width:

$V_H = V_{UT} - V_{LT} = V_{cc}/3$

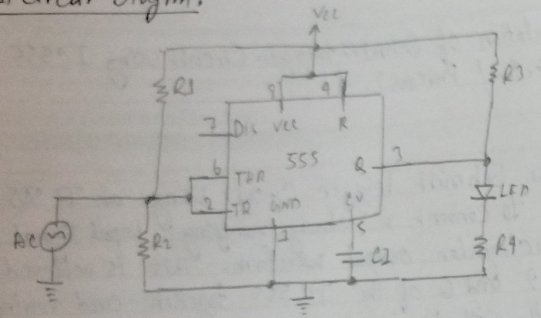
Component Required :-

SN	Name	Specification	Quantity
1	Timer IC	555	1
2	Breadboard	Small	1
3	DC Power Supply	10V	1
4	function Generator	1KHz 2v	1
5	Oscilloscope	–	1
6	Multimeter	Digital	1
7	Resistor	–	required
8	Capacitor	–	required

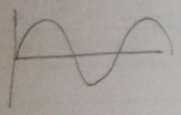
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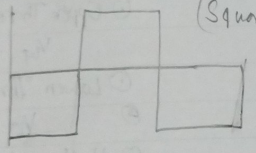
Circuit Diagram:-

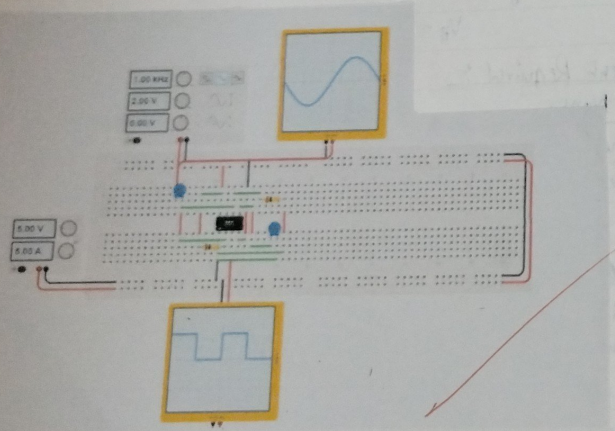


Input:-



Output:-





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Observation Table:-

SN	Vcc (V)	V_{UT} (Theoretical)	V_{LT} (Theoretical)	V_{UT} (Measured) (V)	V_{LT} (Measured) (V)
1	5	3.33	1.67	3.35	1.65
2	9	6.00	3.00	6.02	2.98
3	12	8.00	4.00	8.05	3.95
4	15	10.00	5.00	10.08	4.92

Calculation:-

V_{UT} (Theoretical) = (2/3) × Vcc = (2/3) × 5 = 3.33V

V_{LT} (Theoretical) = (1/3) × Vcc = (1/3) × 5 = 1.67V.

Hysteresis, $V_H = V_{UT} - V_{LT}$

= 3.30 − 1.67

= 1.66V.

Conclusion:-

The Simulation of the Schmitt Trigger circuit using the IC 555 timer was Successfully Completed. The experiment demonstrates that tying pins 2 and 6 together allows the timer to function as a Comparator with build in hysthersis. By analysing the Observation table across various power Supply voltages.

It is evident that the theoretical upper and lower threshold voltages calculated using (2/3) Vcc and (1/3) Vcc closely match the practically measured threshold voltages.

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